



March 15, 2025

Faisal D'Souza
National Coordination Office
Networking & Information Technology Research & Development
National Science Foundation
2415 Eisenhower Avenue
Alexandria, VA 22314

Via email to: [REDACTED]

Re: ***Request for Information on the Development of an AI Action Plan***

Dear Mr. D'Souza:

I am the co-founder and CEO of Compute Exchange,¹ the world's first open exchange for AI compute, the raw computational infrastructure that powers AI by training and running AI models. We appreciate the opportunity to recommend priority policy actions for inclusion in the Trump Administration's AI Action Plan.²

The Problem: AI is Killing Innovation

AI is killing innovation. That has to change, and the Trump Administration can play a critical role in saving innovation and, with it, America's AI dominance.

It sounds like a contradiction but I'm not suggesting that AI itself is doing the damage. Rather, it's the exponentially growing demand for compute and the lack of equitable access to that essential resource.

In other words, no (or insufficient) compute equals no AI-enabled innovation for far too many organizations, large and small. As Vice President Vance said last month at the AI

¹ <https://compute.exchange/about/>

² This document is approved for public dissemination. The document contains no business-proprietary or confidential information. Document contents may be reused by the government in developing the AI Action Plan and associated documents without attribution.

Action Summit in Paris, “the computing power [the AI] stack requires is integral to advancing AI technology.”³

Innovation wasn’t always this gated – and that was a really good thing. For the Internet, letting the free and fair market grow through openness and standardization rapidly transformed the world and made global connectivity possible. The secret wasn’t about control, it was about removing constraints.

But now, as we move deeper into the age of machine learning, AI innovation is choking on its core complements: data and compute. Innovation requires vast computational and data resources. That compute, the fuel that powers this new era, isn’t just expensive; its capacity is constrained and it’s concentrated under the control of the hands of a few very well-resourced organizations. And even those select few suffer from capacity constraints.⁴ In this world – where cost, capacity and control are constrained – innovation gasps.

Despite the recent falling cost of the Graphics Processing Units (GPUs) that are core to AI compute, the cost of access will continue to rise. Researchers from Stanford University and nonprofit research institute Epoch AI have found that the cost of the computational power required to train AI models is doubling every nine months.⁵ If AI proliferates as many predict, supporting every working age person globally (age 15-64) with one H100 GPU (which consumes 1.2 kW of power) would cost nearly \$100 trillion at today’s prices. To put this in perspective for America, the \$500 billion Stargate Project announced the day after President Trump’s inauguration corresponds to just 10% of the over \$5 trillion needed in the U.S. alone in the coming years.

³ Remarks by the Vice President at the Artificial Intelligence Action Summit in Paris, France (Feb. 11, 2025), <https://www.presidency.ucsb.edu/documents/remarks-the-vice-president-the-artificial-intelligence-action-summit-paris-france>.

⁴ See, e.g., CEO Remarks, Google Q4 2024 Earnings Call, <https://blog.google/inside-google/message-ceo/alphabet-earnings-q4-2024/#ai-infrastructure>; Microsoft FY25 Second Quarter Earnings Conference Call, <https://view.officeapps.live.com/op/view.aspx?src=https://cdn-dynmedia-1.microsoft.com/is/content/microsoftcorp/TranscriptFY25Q2>.

⁵ See Epoch AI, How Much Does It Cost to Train Frontier AI Models? (Jan. 13, 2025), <https://epoch.ai/blog/how-much-does-it-cost-to-train-frontier-ai-models>; Stanford Institute for Human-Centered AI (HAI) 2024 AI Index Report, <https://aiindex.stanford.edu/report/>.

The Solution: Increasing Compute Access Through Private Sector Initiatives With Targeted Government Support

Addressing the cost, capacity, and control constraints of the current compute marketplace requires both (1) increasing the amount of available compute and (2) creating a transparent and liquid system for purchasing and selling compute. For innovation to thrive, buyers need to access the amount of compute they need when and where they need it, while providers are able to monetize unused capacity efficiently.

We recommend that the Administration continue its historic AI leadership by including the following four policy priorities in its AI Action Plan:

1. Create a Strategic Compute Reserve and strengthen restrictions on export and foreign use of AI infrastructure to increase national security
2. Establish government-supported public compute resources to provide compute access to researchers and small businesses
3. Make a federal financial commitment to AI infrastructure development to which small and medium-size businesses have access
4. Facilitate the development of voluntary industry standards necessary for a free and fair market for compute

Policy Recommendations

1. Create a Strategic Compute Reserve and strengthen restrictions on export and foreign use of AI infrastructure to increase national security

The United States should establish a Strategic Compute Reserve — a strategic stockpile of AI compute resources to safeguard our AI security — just as it maintains the Strategic Petroleum Reserve⁶ to safeguard energy security. The size of the compute economy is on track to surpass the approximately \$3 trillion oil industry⁷ in the near future so, just as energy independence has been a cornerstone of America's economic and geopolitical strength, we must now treat compute sovereignty with the same level of urgency. A Strategic Compute Reserve would ensure the government has access to critical AI capabilities in times of crisis, protect against supply chain disruptions, and foster domestic innovation.

⁶ U.S. Department of Energy, Strategic Petroleum Reserve (accessed Mar. 15, 2025), <https://www.energy.gov/ceser/strategic-petroleum-reserve>.

⁷ See, e.g., U.S. Energy Information Administration, Short-Term Energy Outlook (Mar. 11, 2025), https://www.eia.gov/outlooks/steo/report/global_oil.php; International Energy Agency, Oil Market Report (Mar. 2025), <https://www.iea.org/reports/oil-market-report-march-2025>.

National security also dictates a careful evaluation of the risks presented by certain foreign countries and actors obtaining American AI infrastructure. We recommend imposing reasonable limits, monitoring, and controls on the export of AI infrastructure to foreign actors posing a national security risk. For example, a limit on the total volume of compute would be better tailored to address concerns about models requiring substantial compute than a limit on the number of chips independent from the actual compute power purchased. In addition, a combination of supply chain, cloud software, and hardware monitoring could provide insights on potentially concerning compute usage. Importantly, a neutral exchange for compute and “one-stop shop” for compute buying and selling such as Compute Exchange is essential to administer any such restrictions and processes transparently and compliantly for the entire market.

National competitiveness, which is necessary for our national security, requires an evaluation of the risks that, if foreign AI developers cannot access American technology, they will be forced to develop and sell their own technology which may put the U.S. at a competitive disadvantage. Accordingly, we caution against an outright ban or actions that may result in American providers being priced out of foreign markets except where there is an unbearable national security risk as to the particular country or entity. For example, we encourage the Administration to repeal the chip embargo on China and, instead, strengthen restrictions on China's access to certain semiconductor equipment necessary for chip manufacture.

2. Establish government-supported public compute resources to provide compute access to researchers and small businesses

Many who want to develop AI applications, including academic and government researchers and startups, are unable to access or afford compute on the current private market.⁸

The Administration should prioritize establishing government-backed computing infrastructures that provide researchers, startups, and public institutions affordable or free access to high-performance AI resources provided by both public and private entities. By lowering the cost barrier to cutting-edge AI development, these initiatives will foster innovation, support essential academic research, and enable smaller enterprises to compete in AI-driven industries.

⁸ See, e.g., NAIRR Task Force, Strengthening and Democratizing the U.S. Artificial Intelligence Innovation Ecosystem (Jan. 2023), <https://www.ai.gov/wp-content/uploads/2023/01/NAIRR-TF-Final-Report-2023.pdf>

In addition, providing subsidized compute will help ensure that innovators in other countries, many of which have or are developing their own government-supported compute clouds,⁹ do not have an unnecessary advantage over American innovators.

We urge the Administration to expand Cloudbank, the National Science Foundation-funded, university-operated, and private sector supported entity that helps the computer science community access and use public clouds for research and education,¹⁰ and increase its support for university-based programs such as the Texas Advanced Computing Center.¹¹ We also recommend that the Administration work with Congress to authorize and appropriate funds for a comprehensive National Artificial Intelligence Research Resource (NAIRR), which is currently in a two year pilot.¹² NAIRR would establish a shared national research infrastructure to connect U.S. researchers to trustworthy AI resources, including compute from both public and private sector providers. Similarly, we urge the Administration to support the authorization and appropriation of funds for the Frontiers in AI for Science, Security and Technology (FASST) program, which includes the creation of public-private partnerships around compute used by and for the Department of Energy.¹³

3. Make a federal financial commitment to AI infrastructure development to which small and medium-size businesses have access

We encourage the Administration to support, and promote open access to and resource sharing in, private initiatives such as the Stargate Project. As noted above, however, even the Stargate Project's \$500 billion investment would provide only a small percentage of the investment likely needed to ensure adequate computational power for American innovators. Public-private partnerships and direct federal investment need to help fill the gap.

⁹ See, e.g., UK Dept. of Science, Innovation & Technology, AI Opportunities Action Plan (Jan. 13, 2025), <https://www.gov.uk/government/publications/ai-opportunities-action-plan/ai-opportunities-action-plan>; Innovation, Science and Economic Development Canada, Canadian Sovereign AI Compute Strategy (accessed Mar. 15, 2025), <https://ised-isde.canada.ca/site/ised/en/canadian-sovereign-ai-compute-strategy>; Charlotte Trueman, Data Center Dynamics, "Japanese government to fund \$470m AI supercomputer" (Apr. 19, 2024), <https://www.datacenterdynamics.com/en/news/japanese-government-to-fund-new-ai-supercomputer/>.

¹⁰ <https://www.cloudbank.org/>

¹¹ <https://tacc.utexas.edu/>

¹² <https://nairrpilot.org/>

¹³ <https://www.energy.gov/fasst>

Importantly, the Administration should ensure that enterprises of all sizes, including small and medium-sized enterprises, are encouraged to apply to participate as partners and recipients of federal funding so that the public opportunities do not mirror the hyperconcentration of the private sector.

4. Facilitate the development of voluntary industry standards necessary for a free and fair market for compute

While the federal government should make direct investments in AI infrastructure to expand compute capacity and improve access, the private sector must lead in developing a sustainable, economically viable solution to ensure that every American innovator has access. For example, Compute Exchange is an open, neutral platform where compute buyers and sellers meet on equal terms, with transparent market-driven pricing where buyers can buy only what they need and sell compute that they do not use.

In order to reduce the complexity and cost of purchasing compute and increase compute utilization rates, which some estimate are well under 50% (wasting billions of dollars),¹⁴ it's important that AI users are able not only to buy compute in a fair and free market but also can migrate their workloads across different compute providers.

We urge the Administration to encourage the development of voluntary standards that would eliminate operational barriers to buying or using compute from different providers. These would include (1) no fees for switching or moving data from one data processing services provider to another, or for data egress from a provider, and (2) interoperability of compute, i.e., being able to use compute purchased from multiple providers/sources at the same time and over time.

We recommend that the Administration add as criteria for procurement and federal investment that compute providers must not charge the government for data egress or transfer, and must not impose any contractual or technical restrictions on the government's use of compute from other sources.

¹⁴ See Uptime Institute Global Data Center Survey 2024, <https://datacenter.uptimeinstitute.com/rs/711-RIA-145/images/2024.GlobalDataCenterSurvey.Report.pdf?version=0>.

Thank you for considering Compute Exchange's views. We look forward to working with the Administration on this important effort.

Sincerely,

Simeon Bochev
Chief Executive Officer
Compute Exchange